Topology-Aware Differentiable Triangle-Soup Reconstruction via Persistent Homology

Anonymous 3DV submission

Paper ID *****

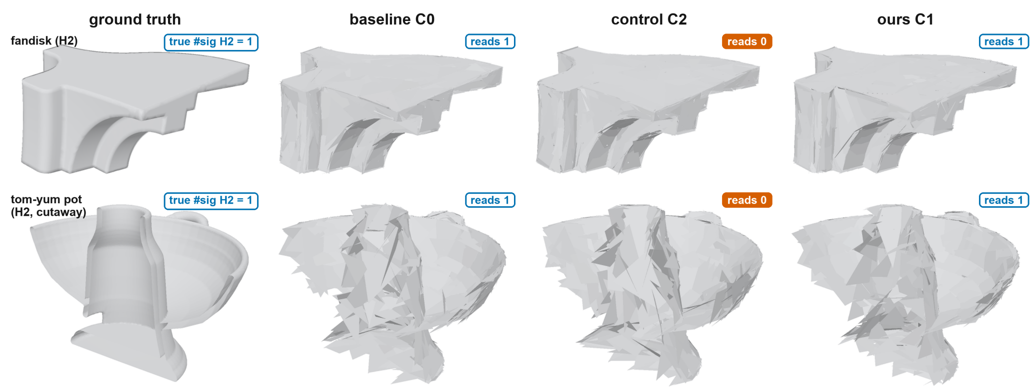

Figure 1. Geometry cannot carry the verdict — measurement can. Final reconstructions (seed 0), identical budgets: ground truth; photometric baseline (C0); a norm-matched non-topological repulsion control through the same gradient channel (C2); the persistence loss (C1, ours), at equal-or-better Chamfer on every row (§5.2). A topological error need not be visible in a render, so each cell is badged with its measured significant-feature count (blue outline = correct, filled orange = wrong; identical across seeds): the control erases both enclosed voids — and collapses a loop on bob (suppl. Fig. S8) — while the loss reads correct everywhere and cuts tail error 2.1–10.4× (Tables 1–2).

Abstract

Differentiable triangle-soup reconstruction inherits a blind spot from its objective: photometric and geometric losses cannot see topology, so a reconstruction with a collapsed loop or a punctured enclosed void can score exactly as well as a correct one (on Chamfer-equal probes the diagrams differ 35–40× in bottleneck distance). The implicit standard remedy — steer where the resampler spends its triangle budget — does not repair this: in a controlled study of that allocation channel, a topology-informed prior is largely matched by an equally wide random one, and no prior shape repairs loops. We therefore move topology into the objective: a differentiable persistence term compares the evolving surface's diagram, measured on live surface samples, to a fixed target; gradients flow through a pair-frozen backward re-expressing matched birth/death simplices as closed-form circumradii (exact for Gabriel simplices, verified per pairing), plus a recruitment term restoring the gradient optimal matching provably lacks when a target feature is missing; one ratio knob calibrates the loss against the photometric gradient, no curriculum needed. Every claim passes a channel-controlled verdict: the loss must beat a norm-matched non-topological control through the identical gradient channel, at Chamfer parity. Under that rule the loss is topology-specific for enclosed voids (4.0–7.9× lower error) and — the class every allocation prior failed — for loops (2.3×, zero phantom handles, while the control collapses one); loss and prior compose; component counts (H0) are an honest null, matched by the generic control. All verdicts replicate without per-shape tuning on four external genus-known meshes (2.1–10.4×; one a raw artist mesh, ingested and certified), degrading gracefully under sensor noise. Scope: all evidence is synthetic and single-machine — closed surfaces plus a first open-surface probe — with the target diagram known in advance, the regime where correction can be measured exactly; real scans and blind capture are future work. Prescription: correct topology in the loss, allocate wide, combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [46], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff while the bottleneck distance between persistence diagrams separates them 35–40× on probes whose Chamfer is equal by construction (§2; suppl. Table S1). Fig. 1 previews the consequence: identical budgets, equal-or-better Chamfer, topology standing or falling with the objective alone.

A decade of connectivity-free reconstruction — point splats, Gaussians, triangle soups — has treated topology errors as symptoms of geometric under-provision: represent the surface better, or allocate the budget more wisely, and topology should follow. The allocation half is directly testable, and we tested it first, biasing where the resampler spends triangles in a controlled prior study (suppl. §A). It fails: a topological prior helps enclosed voids, but any equally-wide non-topological prior reproduces most of the gain, and no prior shape repairs loops — concentrating manufactures phantom handles, spreading never beats a random control.

The null result relocates the root cause: not where triangles are spent, but what the objective measures — a photometric loss is a local, pixel-space signal; a loop or an enclosed void is a global, categorical property of the surface. This paper therefore moves topology into the objective: a differentiable persistence term in the training loss, so gradients act directly on features' birth/death values rather than on where triangles respawn.

Three research hypotheses structure the study, each tested by a pre-designed condition:

1. RH1 (diagnosis): topology errors originate in the training objective, not in primitive allocation — tested by the allocation study above.
2. RH2 (specificity): a persistence term improves topology in ways no norm-matched non-topological regularizer through the identical channel can reproduce — the control C2 (§3.5).
3. RH3 (complementarity): objective and allocation are complementary channels, not substitutes — the composition C5.

Concretely, we contribute a matched persistence loss with a pair-frozen backward (§3): exact persistence forward (GUDHI alpha complex), backward through closed-form circumradii of live surface samples — Gabriel-exact for 100% of significant pairs in practice; a recruitment term for missing features: optimal matching yields exactly zero gradient when a target feature has no live counterpart, recruitment attaches the nearest live bar instead, and removing it collapses the loop-class win to baseline (§5); one calibrated knob: λ set by a gradient-norm ratio ρ, flat over a 10× range, no curriculum needed — an ablation, not an assumption; and channel-controlled evaluation: every claim tested against a norm-matched non-topological control through the identical channel (a repulsion on the same surface samples) — the allocation study's discipline, inherited.

The loss proves topology-specific on the class no allocation prior ever won — loops, with zero phantom handles — confirming RH2; the two channels compose (RH3); and the H0 component class is matched by the generic control, sharpening RH1: topological guidance earns its keep exactly where geometry and topology decouple.

2. Related work

A decade of gradient-based reconstruction: four positions on connectivity. OpenDR [32], neural mesh rendering [26], SoftRas [31], DIB-R [8] and nvdiffrast [28] established gradient flow from pixels to geometry; reconstruction work since splits on what that geometry should be. (i) Implicit fields — learned occupancy and signed distance [34, 39], volume-rendered from NeRF [35] through NeuS [49] to Neuralangelo [30] — are watertight and topologically consistent by construction (the strongest counter-position to explicit soups), at dense-field-evaluation and post-hoc extraction cost. (ii) Fixed-connectivity templates deformed per scene [19, 48] keep manifoldness but freeze topology at the template's genus. (iii) Differentiable iso-surfacing re-extracts the mesh from a volumetric scaffold at every step [36, 43, 44] (Shape-as-Points bridges point soups to watertight implicits [40]) — topological freedom at grid cost. (iv) Connectivity-free primitive soups — point splatting [50], patch atlases [17], Gaussians [27] and their surface-accurate variants [18, 23], triangle splatting [6, 20], and DiffSoup [46] — are the fastest and the easiest to budget, but even their surface-minded members regularize geometry (alignment, flatness, connectivity forces), never a measured topology. (i) and (iii) solve topology representationally; (iv) had not addressed it. Cut by optimization philosophy rather than representation, the decade collapses to two occupied families — geometry-oriented objectives (every method above) and sampling-oriented steering of where a fixed objective spends primitives (below) — and an empty third: topology-oriented objectives that measure the property and differentiate through it (suppl. Table S6). This paper fills the empty family at the soup's budget and speed. We build on DiffSoup, whose in-loop prune/respawn resampler is exactly the mechanism our allocation-channel study (suppl. §A) steered with importance priors; here the resampler stays untouched and the objective changes. Vertex-domain optimizers (large-steps preconditioning [37]) are complementary: they change how vertices move, we change why.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper's problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal by construction (0.916/0.628/0.977% per pair), the discriminating-dimension bottleneck separates correct from broken topology by 35–40× — thin handle ≈0 vs .0302, bridged merge .0014 vs .0574, punctured void .0040 vs .1385 — and Hausdorff$_{95}$ even prefers the wrong candidate in two of three cases (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck 33–53% below baseline — but a width-matched random control recovers most of that gain, leaving a small, shape-dependent residual: the prior's value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the non-topological wide control (.0267 vs the spread prior's .0316), and the concentrated field manufactures phantom handles (4.4 significant loops vs the true 2). The composition condition C5 reuses that study's strongest field (B4; suppl. §A).

Persistence in learning objectives. Persistent homology [13] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [5], for topology-preserving image segmentation [9, 22], point-cloud and shape optimization [4, 41], 3D volume prediction [47], and diagram-metric optimization with convergence analysis [7]; barcode-valued maps have since been given a general differential framework [29], and the per-step cost of persistence-guided descent attacked directly [38]. Nearest our setting, concurrent work brings persistence to reconstruction at other entry points: an image-space barcode term for Gaussian splatting [45], homology-guided camera placement [15] or a genus-matched template [14] in mesh inverse rendering, and a connected-component constraint on implicits fitted to point clouds [24]. Partition our setting's three requirements — (i) a topological quantity of the evolving surface measured differentiably in the objective, (ii) image-based training through a renderer, (iii) a fixed primitive budget — and each neighbour satisfies at most one (suppl. §D.7): the image-space barcode term never measures the surface (i); the point-cloud pipeline forgoes renderer and budget (ii, iii); the template line assumes the genus we measure (i). A head-to-head benchmark is therefore ill-posed (inputs, outputs, and cost regimes differ); our evaluation instead pits the loss against norm-matched controls through its own channel (§4). Our backward pass is in this family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex's birth/death as a closed-form circumradius of live surface samples — with three pipeline-specific ingredients. (i) Alpha-complex pairing over $\alpha\times$area surface samples of a live triangle soup — not image grids or raw clouds — couples the diagram to the geometry actually being trained. (ii) A Gabriel-exactness gate: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — certifying the frozen gradient rather than assuming it. (iii) Recruitment for missing features: optimal partial matching has zero gradient exactly when a target feature is entirely absent — the worst failure mode — so we trade metric fidelity for a restored descent direction (§3), quantifying both its pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [2, 25, 42], and budgeted remeshing has a long lineage [1, 3, 11, 16, 21]; our contribution is orthogonal — the budget mechanics stay DiffSoup's, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary $+\,0.8\,L_1 + 0.2\,\text{SSIM}$) [46], which we adopt unchanged: every ablation below isolates the topology channel, never photometric tuning. We add

$$L \;=\; L_{\text{photo}} \;+\; \lambda(t)\, L_{\text{topo}}(V), \tag{1}$$

where V are the soup's vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed target bundle $\mathcal{T}$: the significant finite bars of the ground-truth shape's diagram, computed once at the sampling density M the loss will see (alpha values shift with spacing: density matching is a correctness requirement), normalized by the target's bounding-box diagonal, and locked for all live evaluations. Throughout, a bar is significant iff its lifetime exceeds the measurement floor $6\,r_{\text{med}}(M)$ (r_{med} = median nearest-neighbour sample spacing; derived and validated in §6).

3.2. Live diagram and pair-frozen backward

At each refresh we draw M surface samples (M=2048 unless the measurement-floor rule of §4 dictates a denser bundle) $X_j = \sum_i w_{ij}\, V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha)\times$area-weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X [12], computed exactly with GUDHI [33], yields persistence pairs (birth simplex, death simplex) per dimension (pipeline: suppl. Fig. S9). For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar's coordinates are closed-form functions of a few sample positions, all exactly differentiable:

$$R_{\text{edge}} = \tfrac{1}{2}\|p_1 - p_0\|, \qquad R_{\text{tri}} = \frac{\|u\|\,\|v\|\,\|u-v\|}{2\,\|u \times v\|}, \qquad R_{\text{tet}} = \|y\| \quad \text{with} \quad A\,y = \tfrac{1}{2}\,\mathrm{diag}\big(AA^{\top}\big), \tag{2}$$

where $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the rows $p_i - p_0$ (the circumcenter is $p_0 + y$). Edges carry H0 deaths and H1 births, triangles H1 deaths and H2 births, tetrahedra H2 deaths. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); a pair failing this Gabriel test is detached — it keeps its (correct) value in the loss but propagates no gradient through that simplex, so an exactness failure can never inject a wrong gradient. In practice the gate never fires: 100% of significant pairs are Gabriel-exact on all six shapes' clean clouds, and the logged per-refresh failure count is zero at every one of the 5,500+ logged refreshes (including all 2,400 noisy ones, §5) — a constant 100% pass rate, not an aggregate. Stronger fallbacks (dropping pairs, subgradients, locally raising M) remain available. The pairing is treated as locally constant — the standard pair-frozen device [5, 7] — refreshed every K=10 steps and, necessarily, after every resampling event (which replaces the vertex and face tensors wholesale); K=10 is an amortization choice, not a tuned knob — it bounds refresh cost (174 ms each; the fixed ≈46 s of §5) while the pairing stays local.

3.3. Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j)\ \text{matched}} \big[(b_i - b_j^*)^2 + (d_i - d_j^*)^2\big] + \sum_{i\ \text{unmatched}} \Big(\tfrac{d_i - b_i}{2}\Big)^2 + L_{\text{recruit}}. \tag{3}$$

The first term pulls matched features to their target coordinates; the second pushes spurious features to the diagonal. The third exists because optimal matching has a blind spot that is provable, not merely observed:

Lemma 1 If a target bar $t \in \mathcal{T}$ is unmatched under the optimal partial matching, then L_{topo} without L_{recruit} is locally independent of t: no component of its gradient in V acts to create the missing feature.

Proof sketch. The unmatched target's augmented cost is its diagonal penalty $\big((d_t^* - b_t^*)/2\big)^2$, constant in the live coordinates; every V-dependent term of Eq. 3 references live bars only, and the matching is locally constant in V (the pair-frozen regime) — no gradient term is directed at t. □

Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target bars, processed in decreasing persistence order, and let the candidate pool $\mathcal{A}$ hold every live bar not claimed by the matching — unmatched significant bars and, crucially, sub-threshold bars, which matching never sees. Each missing target greedily claims its nearest pool bar, without replacement, under the same squared birth–death metric as the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} \big[(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2\big], \qquad k(j) = \operatorname*{arg\,min}_{k \in \mathcal{A}_j} \big[(b_k - b_j^*)^2 + (d_k - d_j^*)^2\big], \tag{4}$$

where $\mathcal{A}_j$ is the pool remaining after earlier recruitments; recruited bars are excluded from the diagonal term and back-propagate through the same circumradius machinery. This deliberately departs from a metric-faithful W_2; §5 probes its known pathology (location-blindness) directly, and the C6 ablation shows the term is load-bearing in training. Greedy rather than optimal assignment is deliberate: in every logged regime the missing-target set is a single bar (torus: one unreached target at every refresh, every seed; §5), where the two coincide; when they diverge, decreasing-persistence order spends the pool on the most significant features first.

3.4. Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho \, \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless (ρ=0.1 throughout); the tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, a constant-$\lambda$ ablation matches the ramped version, and alternative ramp windows land within seed noise (suppl. Table S8) — calibration, not scheduling, makes the knob safe.

3.5. Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: + a short-range repulsion on the same frozen samples, calibrated with the same ρ — the norm-matched control, the RH2 test (C2g: a gentler radius). C3: C1 without the ramp. (C4, a curvature-weighted variant, is deferred to follow-up work; the numbering gap keeps condition tags comparable across studies.) C5: C1 + the best prior of our allocation study (the spread topological field B4; suppl. §A) — the RH3 test. C6: C1 with the recruitment term removed (pure matching + diagonal, all else identical) — does recruitment matter in-loop, or only for cold-start repair?

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout ⇒ 21–63 training views): sphere and cube (void class H2, β=(1, 0, 1), budget N=1200), torus (loop class H1, (1, 2, 1), N=700), two spheres (component class H0, (2, 0, 2), N=700), and genus-2 double torus as a stretch case ((1, 4, 1), N=2000). Budgets are the allocation study's headroom points (suppl. §A) — tight enough that baseline topology is imperfect. Four external, genus-known meshes probe generality beyond the analytic family (§5.2). Three are watertight benchmarks — spot (genus 0) and bob (genus 1) from the Crane model repository (CC0), and the fandisk CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px) at unchanged class budgets; a seed-0 C0 pilot confirms baseline imperfection (tails .043–.054, in the analytic band). The fourth, the tom-yum pot (Fig. 1, bottom row; aluminium showcase: suppl. Fig. S7), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup (9 overlapping open shells, 232 non-manifold junction edges) with no trustworthy ground-truth topology — the same defect that excluded ShapeNet. An ingest-and-certify pass (exact weld, offset-solidification into a thin closed shell, then certification by measurement: exact simplicial homology, watertightness cross-check, bit-identical rebuilds) yields one body of genus 3, β=(1, 6, 1) exact, rendered and budgeted as a void-class shape (N=1200, 48 views, 200 px). This ingest pipeline is a test fixture here, not a claimed contribution. Where the benchmark trio's topology is known by construction, the pot's is known by certificate — what real scans will need (mechanics and challenge inventory: suppl. §B, Table S4). The pot also exercises the floor rule of §6 prospectively: every H1 loop of the certified shell sits below the $6\,r_{\text{med}}$ floor at every working M, while the narrow flue mouth caps at small α — an H2 void on the pot axis, first clearing the floor at M=8192 (margin 1.20–1.23×, five sampling seeds). The pre-registered rule builds the bundle there (the study's only protocol deviation; the loss samples at the same density): exactly one significant bar, stable to re-sampling ($\sim$$10^{-5}$).

Protocol. All conditions share one implementation: C0 is the unmodified DiffSoup trainer; the loss enters as one additive term, and disabled it executes none of the added code (no renderer or resampler edits). 2,500 steps; resampling every 100 steps (untouched); ρ=0.1, ramp 0.2:0.5, refresh K=10, M=2048; decisive shapes (sphere, torus) and the double torus run 5 seeds (the former with the full condition set C0/C1/C2/C3/C5 plus the recruitment ablation C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2). (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss plan sees with i.i.d. Gaussian noise at every refresh, σ = 0.5%/1% of the target diagonal (≈0.5/1.0 median sample spacings at M=2048): the plan decides on the noisy observation, the pulls apply to the true live positions. The torus restricts the loss to H1 (its own void bar is sub-threshold at M=2048 — a loss must not act on features it cannot reliably measure); the double torus likewise targets its two measurable loops of four. The generality wave runs C0/C1/C2, three seeds; bob (fat tube) has both loops and void significant at M=2048 and runs the full bundle; the pot runs at the floor-rule density above.

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape's discriminating dimension, tail-averaged over the last three trajectory dumps (steps 2,300–2,500), seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Reporting policy: mean±sd over seeds; Welch σ only where both arms ran five seeds; three-seed verdicts rest on disjoint per-seed ranges (suppl. Table S7). Verdict rule — this study's core evaluation principle: C1 must beat C0 and every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. Chamfer parity, formally: the passing arm's seed-mean

Chamfer must satisfy $\mathrm{Ch}(\mathrm{C1}) \leq 1.15\,\mathrm{Ch}(\mathrm{C0})$; the 15% headroom absorbs seed variance and nondeterminism (§7) and never did any work — every passing arm's Chamfer was at or below baseline.

5. Results

A gate first: the loss alone (no photometric term, Adam on point positions) repairs all four controlled probes — a punctured sphere's void closes (8.8×), a spurious handle dies at the diagonal, mis-spaced components separate (1229×), and a bridged merge with no significant live H0 bar — the case of Lemma 1 — is severed (450×, $\beta_0\,1{\to}2$); the predicted location-blind pathology did not materialize (suppl. §C, Fig. S5) — sanity toys on raw clouds, not the image pipeline; multipliers not comparable to training results.

5.1. The C-matrix

Table 1 reports tail bottleneck (mean±sd over seeds) in each shape's discriminating dimension; Chamfer in the text where it matters; the full training trajectories for the two decisive shapes are in suppl. Fig. S6 (line = seed mean, band = seed min–max).

Voids (H2): topology-specific, large. Sphere 4.0× (16.2σ Welch vs C0, five seeds), cube 7.9× (three seeds, per the reporting policy: ranges disjoint, C1 .0064–.0086 vs C0 .0551–.0636), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is destructive, not merely weaker: it pins the void's death at its repulsion equilibrium (2.2× worse, identical across seeds) and erases the cube's void outright; no tested strength reproduces any gain — C1 < C0 < C2g < C2 (§7; per-seed spread: suppl. Fig. S3; mechanism: §6).

Loops (H1): the channel's distinguishing result. No prior shape won this class (suppl. §A). The loss cuts torus H1 error 2.3× with zero phantom handles in all five seeds (count trajectories flat at the true β_1=2, suppl. Fig. S4), while the control collapses one loop by step ∼800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not (mechanism: §6).

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error 9.5×, but the generic control matches it on the diagram metric (.0008 vs .0007); the topological increment appears only in Chamfer (0.54 vs 1.94). Three channels now agree: β_0 structure is carried by shell count and separation.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 (7.6× vs C0, ≈2× vs C1 alone, ≈4.4× below the prior alone), torus .0133 — with the best sphere Chamfer; its torus Chamfer (.502) trails only C3's (.495). Allocation and value-tuning are complementary, not substitutes.

Stretch case (genus 2): saturated, honestly. Every double-torus arm returns exactly .0264: the two tube loops are unexpressable at feasible budgets — the representation's ceiling, not the channel's (suppl. §B; floors: §6).

Ablations. The ρ-response is flat over a 10× range (suppl. Table S8); C3 (no curriculum) ≈ C1 on both decisive shapes and off the analytic family (bob: C3 .0171±.0026 vs C1 .0208±.0008). Overhead, honestly: the loss adds a fixed ≈46 s per 2,500-step run (refresh 174 ms at M=2048 every 10 steps) — more than the 33–41 s photometric baseline on these deliberately tiny scenes, but the cost scales $O(M \log M)$ per refresh, not with scene or image complexity, so its share of a run whose photometric cost is T is $46/(T{+}46)$ — an arithmetic projection, ≈7% of a ten-minute rendering-dominated run — independent of image count and resolution; no trainable parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal, all else identical) collapses the torus win entirely: .0411±.0038, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer — though the loops still exist (#sig H1 = 2 every seed; suppl. Fig. S6: C6 rides the baseline). The logs locate the mechanism: one torus loop is significant at every refresh, the other chronically below threshold (all five seeds, all 200 refreshes) — recruitment is the sole gradient path to the second loop; on this class it is the mechanism, not a cold-start device. On the sphere recruitment never fires (0 of 200 refreshes, all seeds): C6 is loss-identical to C1, and its tail gap (.0166 vs .0156, 1.5σ) measures pure run-to-run noise (§7).

Sensor-noise stress (C7/C7h). C7 probes the loss's noise response; it is not a scan-realism claim (structured scan noise: §7). With the plan's cloud perturbed at every refresh (σ = 0.5%/1% of the diagonal ≈ 0.5/1.0 median sample spacings, §4), the loss degrades gradually and never harms: torus .0210±.0018 / .0259±.0022 (2.0×/1.6× below baseline, vs C1's 2.3×), sphere .0235±.0009 / .0328±.0013 (2.7×/1.9×, vs 4.0×) — at equal-or-better Chamfer, with the correct significant-feature count in every run (suppl.

Table 1. Tail bottleneck to target (mean±sd over seeds; tail = last three dumps; lower is better); × = reduction vs C0; verdict: C1 < C0 and < every control at Chamfer parity. Seeds: 5 (sphere, torus, double torus — the latter saturated identically), 3 (cube, two spheres). C6 = no recruitment.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior	C6 no-recruit
sphere (H2)	.0623±.0063	.0156±.0013 (4.0×)	.1372 (worse)	.0151±.0004	.0082±.0007 (7.6×)	.0166±.0005
cube (H2)	.0582±.0046	.0074±.0011 (7.9×)	.1346 (β_2 1→0)	—	—	—
torus (H1)	.0424±.0031	.0180±.0016 (2.3×)	.0457 (β_1 2→1)	.0155±.0015	.0133±.0004 (3.2×)	.0411±.0038
two spheres (H0)	.0065±.0008	.0007±.0002 (9.5×)	.0008 (ties C1)	—	—	—
double torus (H1)	.0264±.0000	.0264±.0000 (saturated)	.0264±.0000	—	—	—

Table 2. Generality wave: tail bottleneck (mean±sd, 3 seeds); × = reduction vs C0. All four pass the verdict rule of Table 1. No Welch σ at three seeds (§4): every shape's C0/C1 per-seed ranges are disjoint (suppl. Table S7).

shape (dim)	C0	C1 loss	C2 control
spot (H2)	.0521±.0081	.0237±.0000 (2.2×)	.0652 (β_2 1→0)
bob (H1)	.0426±.0012	.0208±.0008 (2.1×)	.0437 (β_1 2→1)
fandisk (H2)	.0538±.0022	.0052±.0008 (10.4×)	.0571 (β_2 1→0)
tom-yum pot (H2)	.0221±.0003	.0057±.0002 (3.9×)	.0223 (β_2 1→0)

Fig. S6) and zero Gabriel-gate failures in all 2,400 noisy refreshes (further internals: suppl. §B).

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule (§4) — on four external genus-known meshes — spot (0, organic), bob (1, organic), fandisk (0, CAD), the tom-yum pot (3, artist-authored, certified by ingest; §4) — with no per-shape tuning of any kind. All four pass the verdict rule at equal-or-better Chamfer (Table 2; trajectories in suppl. Fig. S2), and the control's failure modes replicate exactly: the void erased on all three H2 shapes (β_2 1→0, all seeds; the pot at 1.36× worse Chamfer), a loop collapsed on the genus-1 shape (β_1 2→1, all seeds).

Four observations. The H1 claim survives the family change: bob halves loop error with zero phantom handles in every seed (rendered: suppl. Fig. S8), replicating torus. Sharp-featured CAD benefits most: fandisk's 10.4× exceeds every analytic-family effect. Complex organic geometry exposes a count-level limit: spot's baseline manufactures a phantom void in two of three seeds (final #sig H2 = $2, 1, 2$); the loss clears two, retains one spurious bar in the third $(1, 2, 1)$ — value repair is reliable, count repair on sub-budget organic detail (legs, horns at N=1200) is not guaranteed; spot's C1 tail pins at .0237 (sd 0), a milder floor of the double-torus kind. And the pipeline reaches raw artist content: the tom-yum pot (Fig. 1; suppl. Fig. S7) extends the wave in input class (a raw Blender export, certified rather than constructed; §4) and observable (an H2 void clearing the floor only at M=8192), both chosen by the pre-registered rule — the dress rehearsal for scan data. The loss cuts the void's value error 3.9× at Chamfer parity (0.99× baseline), with the tightest seed spread of the study (C1 .0056–.0059 vs C0 .0218–.0223; at n=3 a Welch σ would inflate to a meaningless headline — hence ranges), inside the study's 2.1–10.4× span. The control destroys the void (β_2 1→0 in all three seeds) at 1.36× worse Chamfer; its bottleneck pins at the unmatched-target bound (.0223), so the count — not the value gap — is the verdict. Two caveats transfer: the baseline already reads the correct count (a value-accuracy win, like bob), and ground truth is certified, not constructed.

6. Mechanism and discussion

A local signal cannot carry a global property — sparse tugs can. A pixel's photometric gradient reaches only the few vertices rasterizing to it; topology is global and categorical — a wrong genus is not approached smoothly, and the blindness probes quantify the gap (§2) — no geometric proxy stands in. The persistence term supplies the measurement with a usable gradient: each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices — precise but sparse pulls, iterated by re-pairing — the toys show surgery "walking" around a defect a few points at a time; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort along this axis: loops respond to value pressure (H1 win), wide shells to both channels (composition best), components to neither (H0 null).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void's death pins at that equilibrium (identical across seeds) and drags while near-diagonal phantom bars accumulate (suppl. Fig. S1) and thin structures collapse (the cube's β_2, the torus's β_1); the loss has no length scale — it pulls the death coordinate itself. This sharpens the control argument: the same gradient budget through the same channel without topological information is actively harmful, so C1's wins cannot be extra-regularization artifacts.

Measurability at the loss's density. A persistence loss can only enforce features that clear the significance floor at its sampling density M: the torus void and the double torus's two tube loops do not, at M=2048. We treat this as a principle, not a nuisance: the bundle is built (and audited) at the same M, sub-floor features are excluded, and the double torus is evaluated against its two measurable loops — all the reconstruction can express at feasible budgets.

The floor is quantifiable: a bar is significant iff its lifetime exceeds $6\,r_{\mathrm{med}}(M)$ (r_{med} = median nearest-neighbour spacing), and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090→.0028 over M=2048→20000; ratio 3.16 vs $\sqrt{9.77}$=3.13; visualized: suppl. Fig. S10). The double torus's tube loops live at lifetime .045–.046 on the clean surface: at M=2048 that is 0.83–0.85× the floor — invisible by a 15% margin, clearing it only from $M \approx 3\times10^3$ — while at the evaluation density (M=20000, floor .0171) they clear it 3.1×; the binding constraint there is the representation (the soup never expresses the tubes), so raising M alone cannot close the gap — the measurement floor recedes as $M^{-1/2}$, the representation floor is set by the budget N and no M lowers it, and the double torus sits on the second. The torus quantifies the noise interaction: its second loop clears the clean floor by only 1.33× — a margin live-cloud noise consumes, leaving recruitment as its only gradient path (C6), the loss reaching below its own floor (suppl. §C). Suppl. Table S3 tabulates counts across M; the pot ran the rule prospectively (§4), validated in-loop (Table 2); adaptive density is the natural extension, at $O(M \log M)$.

Prescription. Correct topology in the loss; allocate with a wide prior; combine. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7. Limitations

Synthetic, single-machine. All quantitative claims are on synthetic renders with known topology: an analytic family plus four external genus-known meshes (§5.2; one a certified raw artist mesh — the intended path for scans), short of a public benchmark. All surfaces are closed except a first open-surface probe, pre-registered and passed: two bowls cut from the sphere (openings $0.14R/0.50R$, certified disk topology) still read one significant void (the designed rim's H1 never clears the floor); the loss cuts its error 4.2×/4.1× at better-than-baseline Chamfer, ranges disjoint, while the control erases the wide bowl's void in two of three seeds (suppl. §F). C7's Gaussian probe bounds the loss's noise response, not scan realism (real noise brings outliers, missing regions, registration error). Real scans remain future work: their boundaries add noise-born H1 bars (unlike a designed rim); bar-filtering them is the near-term step.

Density floor. The loss cannot see features below the significance floor at its density (torus void; two of four double-torus loops at M=2048) — audited visibly, but a real ceiling. Count-level repair of sub-budget detail is likewise out of scope (spot: the baseline's phantom void is cleared in two of three seeds, not guaranteed away; §5.2).

Fixed target diagram. The target bundle presumes known ground-truth topology (a prior scan or template — not blind capture); template-free designs (total-persistence minimization; a target-genus penalty, our first candidate; class-level diagram priors) are untested.

Control strength. The repulsion control is aggressive (it destroys topology at calibrated magnitude); a gentler variant at half radius is still 1.6× worse (sphere: .0972 vs .0623), so verdicts do not hinge on control tuning.

Nondeterminism. CUDA training is not bit-reproducible: divergence enters at resampling, where borderline prune/split decisions flip under float drift; tail CVs run 7–10% for baseline and loss arms alike, and a loss-identical re-run pair (sphere C6 ≡ C1) landed 6% apart — pure noise. Statements follow §4's reporting policy.

Recruitment guarantees. The zero-gradient pathology it repairs is provable (Lemma 1); recruitment's own self-correction premise is empirical, not proven (suppl. §C) — pathological geometries could defeat it; a failure-case study is future work.

8. Conclusion

We moved topology from the resampler's allocation channel into the training objective: a matched persistence loss with a Gabriel-certified backward and a recruitment term for missing features (load-bearing for loops), every gain beating a norm-matched control at Chamfer parity. The hypotheses hold (RH1: root cause in the objective; RH2: signal specifically topological; RH3: channels compose); verdicts replicate on external meshes, degrading gracefully under noise. Prescription: correct topology in the loss, allocate wide, combine; next: real scans. Reproducibility: CUDA training is not bit-reproducible (§7); code, scenes, target bundles, logs, and seed lists ship on acceptance.

References

[1] Pierre Alliez, David Cohen-Steiner, Olivier Devillers, Bruno Lévy, and Mathieu Desbrun. Anisotropic polygonal remeshing. ACM Transactions on Graphics, 22 (3):485–493, 2003. 3

[2] Marco Attene. A lightweight approach to repairing digitized polygon meshes. The Visual Computer, 26 (11):1393–1406, 2010. 3

[3] Mario Botsch and Leif Kobbelt. A remeshing approach to multiresolution modeling. In Eurographics/ACM SIGGRAPH Symposium on Geometry Processing (SGP), 2004. 3

[4] Rickard Brüel-Gabrielsson, Vignesh Ganapathi-Subramanian, Primoz Skraba, and Leonidas J. Guibas. Topology-aware surface reconstruction for point clouds. Computer Graphics Forum, 39(5): 197–207, 2020. 3

[5] Rickard Brüel Gabrielsson, Bradley J. Nelson, Anjan Dwaraknath, and Primoz Skraba. A topology layer for machine learning. In International Conference on Artificial Intelligence and Statistics (AISTATS), pages 1553–1563, 2020. 3, 4

[6] Nathaniel Burgdorfer and Philippos Mordohai. Radiant triangle soup with soft connectivity forces for 3D reconstruction and novel view synthesis. arXiv:2505.23642, 2025. 2

[7] Mathieu Carrière, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hariprasad Kannan, and Yuhei Umeda. Optimizing persistent homology based functions. In International Conference on Machine Learning (ICML), 2021. 3, 4

[8] Wenzheng Chen, Jun Gao, Huan Ling, Edward J. Smith, Jaakko Lehtinen, Alec Jacobson, and Sanja Fidler. Learning to predict 3D objects with an interpolation-based differentiable renderer. In Advances in Neural Information Processing Systems (NeurIPS), 2019. 2

[9] James R. Clough, Nicholas Byrne, Ilkay Oksuz, Veronika A. Zimmer, Julia A. Schnabel, and Andrew P. King. A topological loss function for deep-learning based image segmentation using persistent homology. IEEE Transactions on Pattern Analysis and Machine Intelligence, 44(12):8766–8778, 2022. 3

[10] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. Discrete & Computational Geometry, 37(1):103–120, 2007. 3

[11] Marion Dunyach, David Vanderhaeghe, Loïc Barthe, and Mario Botsch. Adaptive remeshing for real-time mesh deformation. In Eurographics 2013 — Short Papers, 2013. 3

[12] Herbert Edelsbrunner and Ernst P. Mücke. Three-dimensional alpha shapes. ACM Transactions on Graphics, 13(1):43–72, 1994. 4

[13] Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. Discrete & Computational Geometry, 28(4):511–533, 2002. 3

[14] Xiang Gao, Xinmu Wang, Xiaolong Wu, Jiazhi Li, Jingyu Shi, Yu Guo, Yuanpeng Liu, Xiyun Song, Heather Yu, Zongfang Lin, and Xianfeng David Gu. Inverse rendering for high-genus surface meshes from multi-view images. arXiv:2511.18680, 2025. 3DV 2026. 3

[15] Xiang Gao, Xinmu Wang, Yuanpeng Liu, Yue Wang, Junqi Huang, Wei Chen, and Xianfeng Gu. Inverse rendering for high-genus 3D surface meshes from multi-view images with persistent homology priors. arXiv:2601.12155, 2026. ICASSP 2026. 3

[16] Michael Garland and Paul S. Heckbert. Surface simplification using quadric error metrics. In Proceedings of SIGGRAPH 97, pages 209–216, 1997. 3

[17] Thibault Groueix, Matthew Fisher, Vladimir G. Kim, Bryan C. Russell, and Mathieu Aubry. AtlasNet: A papier-mâché approach to learning 3D surface generation. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2018. 2

[18] Antoine Guédon and Vincent Lepetit. SuGaR: Surface-aligned Gaussian splatting for efficient 3D mesh reconstruction and high-quality mesh rendering. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 5354–5363, 2024. 2

[19] Rana Hanocka, Gal Metzer, Raja Giryes, and Daniel Cohen-Or. Point2Mesh: A self-prior for deformable meshes. ACM Transactions on Graphics, 39(4), 2020. 2

[20] Jan Held, Renaud Vandeghen, Adrien Deliège, Abdullah Hamdi, Silvio Giancola, Anthony Cioppa, Andrea Vedaldi, Bernard Ghanem, Andrea Tagliasacchi, and Marc Van Droogenbroeck. Triangle splatting for real-time radiance field rendering. arXiv:2505.19175, 2025. 2

[21] Hugues Hoppe. Progressive meshes. In Proceedings of SIGGRAPH 96, pages 99–108, 1996. 3

[22] Xiaoling Hu, Fuxin Li, Dimitris Samaras, and Chao Chen. Topology-preserving deep image segmentation. In Advances in Neural Information Processing Systems 32 (NeurIPS), 2019. 3

[23] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2D Gaussian splatting for geometrically accurate radiance fields. In ACM SIGGRAPH 2024 Conference Papers, 2024. 2

[24] Anushrut Jignasu, Ethan Herron, Zhanhong Jiang, Soumik Sarkar, Chinmay Hegde, Baskar Ganapathysubramanian, Aditya Balu, and Adarsh Krishnamurthy. STITCH: Surface reconstruction using implicit neural representations with topology constraints and persistent homology. arXiv:2412.18696, 2024. 3

[25] Tao Ju. Robust repair of polygonal models. ACM Transactions on Graphics, 23(3):888–895, 2004. 3

[26] Hiroharu Kato, Yoshitaka Ushiku, and Tatsuya Harada. Neural 3D mesh renderer. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 3907–3916, 2018. 2

[27] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. ACM Transactions on Graphics, 42(4):139:1–139:14, 2023. 2

[28] Samuli Laine, Janne Hellsten, Tero Karras, Yeongho Seol, Jaakko Lehtinen, and Timo Aila. Modular primitives for high-performance differentiable rendering. ACM Transactions on Graphics, 39(6):194:1–194:14, 2020. 2

[29] Jacob Leygonie, Steve Oudot, and Ulrike Tillmann. A framework for differential calculus on persistence barcodes. Foundations of Computational Mathematics, 22:1069–1131, 2022. 3

[30] Zhaoshuo Li, Thomas Müller, Alex Evans, Russell H. Taylor, Mathias Unberath, Ming-Yu Liu, and Chen-Hsuan Lin. Neuralangelo: High-fidelity neural surface reconstruction. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. 2

[31] Shichen Liu, Tianye Li, Weikai Chen, and Hao Li. Soft rasterizer: A differentiable renderer for image-based 3D reasoning. In IEEE/CVF International Conference on Computer Vision (ICCV), pages 7708–7717, 2019. 2

[32] Matthew M. Loper and Michael J. Black. OpenDR: An approximate differentiable renderer. In European Conference on Computer Vision (ECCV), pages 154–169, 2014. 2

[33] Clément Maria, Jean-Daniel Boissonnat, Marc Glisse, and Mariette Yvinec. The Gudhi library: Simplicial complexes and persistent homology. In International Congress on Mathematical Software (ICMS), pages 167–174, 2014. 4

[34] Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. Occupancy networks: Learning 3D reconstruction in function space. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019. 2

[35] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In European Conference on Computer Vision (ECCV), 2020. 2

[36] Jacob Munkberg, Jon Hasselgren, Tianchang Shen, Jun Gao, Wenzheng Chen, Alex Evans, Thomas Müller, and Sanja Fidler. Extracting triangular 3D models, materials, and lighting from images. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 8280–8290, 2022. 2

[37] Baptiste Nicolet, Alec Jacobson, and Wenzel Jakob. Large steps in inverse rendering of geometry. ACM Transactions on Graphics, 40(6), 2021. 3

[38] Arnur Nigmetov and Dmitriy Morozov. Topological optimization with big steps. Discrete & Computational Geometry, 2024. 3

[39] Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. DeepSDF: Learning continuous signed distance functions for shape representation. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 165–174, 2019. 2

[40] Songyou Peng, Chiyu Jiang, Yiyi Liao, Michael Niemeyer, Marc Pollefeys, and Andreas Geiger. Shape as points: A differentiable Poisson solver. In Advances in Neural Information Processing Systems (NeurIPS), 2021. 2

[41] Adrien Poulenard, Primoz Skraba, and Maks Ovsjanikov. Topological function optimization for continuous shape matching. Computer Graphics Forum, 37 (5):13–25, 2018. 3

[42] Andrei Sharf, Thomas Lewiner, Gil Shklarski, Sivan Toledo, and Daniel Cohen-Or. Interactive topology-aware surface reconstruction. ACM Transactions on Graphics, 26(3):43:1–43:10, 2007. 3

[43] Tianchang Shen, Jun Gao, Kangxue Yin, Ming-Yu Liu, and Sanja Fidler. Deep marching tetrahedra: a hybrid representation for high-resolution 3D shape synthesis. In Advances in Neural Information Processing Systems (NeurIPS), 2021. 2

[44] Tianchang Shen, Jacob Munkberg, Jon Hasselgren, Kangxue Yin, Zian Wang, Wenzheng Chen, Zan Gojcic, Sanja Fidler, Nicholas Sharp, and Jun Gao. Flexible isosurface extraction for gradient-based mesh optimization. ACM Transactions on Graphics, 42(4), 2023. 2

[45] Tianqi Shen, Shaohua Liu, Jiaqi Feng, Ziye Ma, and Ning An. Topology-aware 3D Gaussian splatting: Leveraging persistent homology for optimized structural integrity. In Proceedings of the AAAI Conference on Artificial Intelligence, pages 6823–6832, 2025. 3

[46] Kenji Tojo, Bernd Bickel, and Nobuyuki Umetani. DiffSoup: Direct differentiable rasterization of triangle soup for extreme radiance field simplification. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. arXiv:2603.27151. 2, 3

[47] Dominik J. E. Waibel, Scott Atwell, Matthias Meier, Carsten Marr, and Bastian Rieck. Capturing shape information with multi-scale topological loss terms for 3D reconstruction. In Medical Image Computing and Computer Assisted Intervention (MICCAI), pages 150–159, 2022. 3

[48] Nanyang Wang, Yinda Zhang, Zhuwen Li, Yanwei Fu, Wei Liu, and Yu-Gang Jiang. Pixel2Mesh: Generating 3D mesh models from single RGB images. In European Conference on Computer Vision (ECCV), 2018. 2

[49] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. In Advances in Neural Information Processing Systems (NeurIPS), 2021. 2

[50] Wang Yifan, Felice Serena, Shihao Wu, Cengiz Öztireli, and Olga Sorkine-Hornung. Differentiable

surface splatting for point-based geometry processing. ACM Transactions on Graphics, 38(6), 2019. 2